ASHLEY LIU

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EDUCATION

University of California, San Diego - M.S., Computer Engineering — Expected 12/2026
Selected Coursework: *Machine Learning, Deep Generative Models, NLP, Recommender Systems, AI Agents, Algorithms, CV, Distributed Systems*
Lancaster University & Beijing Jiaotong University - Dual B.S./B.E. in Computer Science, Major GPA 3.76 — 09/2020 - 06/2024

WORK EXPERIENCE

***Software Engineer | AI/ML Intern,* California ISO (CAISO), Folsom, CA** — 06/2026 - 09/2026

- Owned end-to-end development of CTRACC, a NERC PER-005 training and qualification platform built with JavaScript, PHP, SQL Server, and IIS/FastCGI; architected a version-aware relational schema separating stable task identities from revision-specific records, preserving exact version associations across qualification cards, course objectives, learner assignments, requalification, and historical reporting.
- Built an 8-stage deterministic retrieval pipeline for 120+ versioned articles using field-weighted BM25F, four-channel Reciprocal Rank Fusion, concept/alignment scoring, and confidence gating; achieved 91.7% Top-1 accuracy and logged query/candidate scores for ranking diagnostics.
- Designed seven k6 load/concurrency tests across three IIS/FastCGI applications; diagnosed and mitigated an 8.2× p95 latency regression (710ms→5.83s) caused by shared PHP worker contention through SQL Server DMV analysis, process sampling, and worker tuning; integrated Claude Code with Playwright MCP for automated end-to-end HTTP/database validation.

***Software Development Engineer*, Advanced Micro Devices, Boxborough, MA** — 04/2025 - 10/2025

- Built and deployed a Docker-containerized APU/GPU lifecycle-management platform using Spring Boot, REST APIs, MyBatis, MySQL, and Vue 3, modeling asset/request state transitions with RBAC, validation, IP traceability, and ticket/waitlist workflows through NGINX and CI/CD.
- Developed multithreaded C++/Python automation for parallel GPU test execution, result validation, and failure triage across Linux and Windows; maintained KitChecker and QuickEPR in C#/VB.NET to support repeatable hardware and software validation workflows.
- Investigated GPU correctness, driver compatibility, and performance on Linux/Windows, using GPU PerfStudio, Radeon GPU Analyzer, DirectX, Vulkan, and OpenGL to isolate driver/system failures; tracked defects in Jira and supported rollback and failure recovery workflows.

***Technical Consultant*, AX Future Academy, Boston, MA** — 12/2024 - 04/2025

- Built Java/Spring backend services for AI-assisted search and ranking, integrating data-driven web features with cloud deployment workflows.

***Image Processing Engineer Intern*, Zuoyouming Technology Co., Ltd, Hubei, CN** — 07/2023 - 09/2023

- Built PyTorch U-Net segmentation and YOLOv5 detection pipelines over 4,340 UAV orthophotos; implemented tiling, augmentation, Labelme JSON conversion, dataset validation, Dice/cross-entropy loss experiments, and batch inference for water, road, and building extraction.

***Android Remote Control Application Developer Intern,* Hikeen Technology Co., Ltd, Shenzhen, CN** — 07/2022 - 09/2022

- Built a layered Android TV control stack with separate UI, network, and command layers, using UDP broadcast discovery, session management, sequence/ACK/retry logic, and asynchronous sockets for reliable low-latency command delivery and gesture translation over unstable networks.

PROJECT EXPERIENCE

Agentic Chart Reasoning System | Python, CrewAI, Pydantic, Multimodal Evals — 03/2026 - 05/2026

- Architected a multi-agent chart-QA system using Python, CrewAI, and Pydantic-typed ChartDSL to bind chart values to axis/legend categories; routed lookup, comparison, counting, and arithmetic queries to specialist solvers sharing structured evidence through a blackboard.
- Built critique, answer normalization, and prediction-audit stages to check evidence support, paired IDs, missing outputs, and exact-match scores; historical multi-agent predictions reached 70.0% strict EM vs. 36.7% for the GPT-4o-mini baseline on DVQA val_hard.

Hierarchical Chest X-ray Generation | PyTorch, Variational Autoencoders, GANs, Computer Vision — 10/2026 - 11/2026

- Designed a hierarchical VAE from scratch with 128-D global latents and conditional 8×32×32 spatial latent; trained on 4,694 chest X-rays using MSE, SSIM, Sobel loss, KL annealing, and 0.35 local-KL weighting, reducing validation MSE by 31.9% over an MSE-only baseline.
- Fine-tuned the VAE decoder with a spectral-normalized PatchGAN critic while freezing the encoder/priors;used 5% gradient-norm-targeted adversarial weighting, reducing FID 216.52→210.14 (n=522); assessed reconstruction-generation tradeoffs with KID,FID and Inception Score.

Multi-Aspect Review Scoring | NLP / Applied Machine Learning — 11/2025 - 12/2025

- Built a five-aspect DistilBERT regressor with rank-16 LoRA Q/V adapters, a trainable MLP head, and masked loss/metrics for missing labels.
- Designed product-held-out splits and exact-text deduplication; verified LoRA adapter/head/tokenizer export and reload with 13 pipeline checks. Separately, a Ridge baseline reduced macro MSE 12.2% versus training-mean prediction on 3,559 held-out reviews.

PUBLICATIONS AND SELECTED AWARDS

- **Liu, X**.; Tao, H. "Java-Based Development of a Vehicle Speed Recommendation Model for Assisted Driving." IEEE Conference on Machine Learning, Algorithms and Artificial Intelligence (MLAAI), 2023. (Paper ID: MLAAI-2368)
- **1st Prize-**National Univ. Intelligent Vehicle Competition (Vision); Outstanding Graduation Project (Multimodal Lip Recognition, 2023–2024)

SKILLS

Languages: Python, C++, Java, TypeScript/JavaScript, SQL, C#, Go
ML / AI: PyTorch, TensorFlow, Transformers, CUDA, Triton, TensorRT, vLLM, LoRA/PEFT, scikit-learn, OpenCV, RAG, LLM Evaluation
Backend / Data: Spring Boot, Flask, REST APIs, MySQL, SQL Server, MyBatis, Node.js, React, Vue 3, Pydantic
Systems / Infrastructure: Linux, AWS (EC2, Lambda), Docker, Kubernetes, NGINX, HTTP, CI/CD, GPU Profiling, Git, IIS/FastCGI, TCP/UDP